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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

January 2019 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4368-QP-JAN19

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 2
Data Book (Examination)

January 2019

Insert

Instructions
- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice
- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the Question Paper/Answer Book.

4368

QAN 500/8702/2

IB/M/Jan19/E4

Data Sheet for On the farm

Dairy farm

A dairy farm is a farm where cows are kept for their milk.



The amount of milk a cow produces is called the **milk yield**.

This amount is measured in litres.

In the UK, the mean milk yield per cow per year is 7944 litres.

Farmers usually sell the milk for between 30p and 34p per litre.

Keeping pigs

Pigs are usually kept in buildings called **sties**.
Each **sty** should have a covered area to keep the pigs dry.
Each sty should have a gate at one end to allow access to open space.
Pigs like to have space to run around and digging areas to look for food.



A mother pig is called a sow.

Baby pigs are called piglets.

The table shows the minimum area of sty needed for sows and their piglets.

	Total number of pigs in the sty	
	5 or fewer	**6 or more**
Sow	2.48 square metres each	2.25 square metres each
Piglet	1.80 square metres each	1.64 square metres each

Data Sheet for Part-time work

Tram timetables

Queens Road to Piccadilly Gardens					
Trams are at these minutes past each hour	01	13	25	37	49
The journey time from Queens Road to Piccadilly Gardens is 11 minutes					

Piccadilly Gardens to Queens Road					
Trams are at these minutes past each hour	04	16	28	40	52
The journey time from Piccadilly Gardens to Queens Road is 10 minutes					

Bus timetables

Piccadilly Gardens to Trafford Centre								
Piccadilly Gardens	0715	0745	0815	0845	0912	0932	0952	1012
Deansgate Station	0722	0753	0823	0853	0920	0940	1000	1020
Old Trafford	0732	0803	0833	0903	0930	0950	1010	1030
Trafford Centre	0747	0818	0848	0918	0945	1005	1025	1045

Trafford Centre to Piccadilly Gardens								
Trafford Centre	1733	1753	1813	1833	1853	1913	1933	1953
Old Trafford	1748	1808	1828	1848	1908	1928	1948	2008
Deansgate Station	1759	1819	1839	1859	1919	1937	1957	2017
Piccadilly Gardens	1806	1826	1846	1906	1926	1943	2003	2023

The National Minimum Wage

Workers should be paid at least the National Minimum Wage.

This table shows the hourly rates for different ages.

Age (years)	16 to 17	18 to 20	21 to 24	25 or over
Hourly rate	£4.20	£5.90	£7.38	£7.83

The Real Living Wage

In November 2017, a Real Living Wage of £8.75 per hour was recommended for all workers.

END OF DATA

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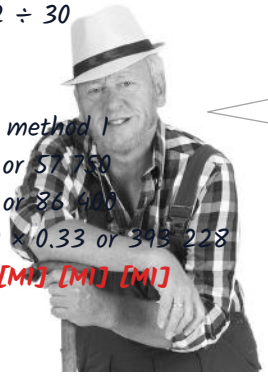
Do not write outside the box

Answer **all** questions in the spaces provided.

1 **On the farm**

There is a **data sheet** for On the farm.

Q1(a) Worked Solution:
652.2
their 652.2 ÷ 30
21.74
Answer
Alternative method 1
385 × 150 or 57 750
48 × 1800 or 86 400
7944 × 150 × 0.33 or 393 228
[M1] [A1] [M1] [M1] [M1]



I am a farmer.
I keep cows and pigs.

Ken

1 (a) One day, Ken records the milk yields, in litres, for 30 of his cows.

Number of litres	Frequency
21.6	5
21.7	15
21.8	5
21.9	3
22.0	2

Work out the mean milk yield for these 30 cows.

[3 marks]

Do not write outside the box

1 (b) In total, Ken has 150 cows.

The mean milk yield per cow is 7944 litres per year.

Ken sells all the milk for 33p per litre.

The table shows the cost of looking after his cows this year.

Milking	£385 per **cow**
Food	£48 per **tonne**
Other costs	£189 550

Ken buys 1800 tonnes of food per year.

He says,

"This year, my profit on the milk from these cows is **more than** £60 000"

Is he correct?
You **must** show your working.

[6 marks]

Do not write outside the box

Ken wants to build a rectangular sty for 1 sow and 7 piglets.

†1 (c) Work out the minimum area of sty he needs to build.

[2 marks]

Check your answer.
Show how you have done your check.

[1 mark]

1 (d) Ken has

12 sections of fencing, each of length 1 metre

one gate, of width 2 metres.

He uses **all** the fencing and the gate to build the rectangular sty.

He does not cut the sections of fencing.

Can he build a sty large enough for 1 sow and 7 piglets?
You **must** show your working.

[4 marks]

16

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2 **Part-time work**

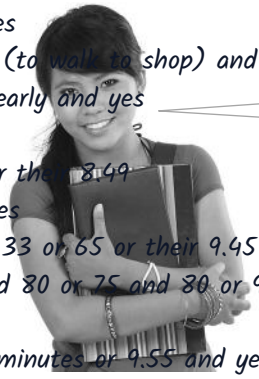
There is a **data sheet** for Part-time work.

Emmie lives near Queens Road.

Q2(a) Worked Solution:

9.55 and yes
15 minutes (to walk to shop) and yes
5 minutes early and yes
Answer
their 9.01 or their 8.49
8.44 and yes
their 32 + 33 or 65 or their 9.45
65 + 10 and 80 or 75 and 80 or 9.55
and yes
8.40 + 75 minutes or 9.55 and yes
9.40 and yes B1ft **Emmie**

3 Wrong tram time but arrives at 9.55



I work part-time in a shop in the Trafford Centre.

Here are the details of Emmie's journey to work.

[B2] [B1] [B2] [B2] [B1] [B1]

5-minute walk from home to the Queens Road tram stop
Get the tram from Queens Road to Piccadilly Gardens
Get the bus from Piccadilly Gardens to the Trafford Centre
10-minute walk from the bus stop at the Trafford Centre to the shop

2 (a) Emmie starts work at 10.00 am

She says,

"If I leave home at 8.40 am I should get to work on time."

Is she correct?

You **must** show your working.

[5 marks]

Do not write outside the box

2 (b) Emmie is 21 years old.

She is paid the National Minimum Wage.

She works for 2 days each week.

On each of the 2 days, she

works for 4 hours

pays £4 in tram fares

pays £3.70 in bus fares.

Emmie pays the fares from her wages.

How much money does she have left from her wages each week?

[4 marks]

2 (c) Emmie changes jobs.

In her new job, she is paid the Real Living Wage.

She works for 3 days each week.

On each of the 3 days she works 4 hours.

She has no travel costs.

Emmie says,

"In my new job, I am better off by **more than** £60 per week."

Is she correct?

You **must** show your working.

[3 marks]

Do not write outside the box

2 (d) Jo works with Emmie.

She is paid a salary of £21 105

She uses these steps to work out the Income Tax she should pay.

Step 1 Subtract £11 850 from her salary

Step 2 Work out 20% of the answer to **Step 1**

The answer to **Step 2** is the Income Tax Jo should pay.

How much should she pay?

[3 marks]

15

Turn over for the next question

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3 Log burner

Rohan wants to put a log burner in his living room.

Q3(a) Worked Solution:

their 75.9 – their
their 370 ÷ 10
38.9
37
alternative or reverse method, e.g.
their 37 × 10 = 370
and
their 370 + 389 = 789
estimation, e.g.
760 ÷ 10 or 76 and 390 ÷ 10 or 39



Here is some information about four log burners.

The log burners are called the Aston, the Brunel, the Chester and the Dover.

76 – 39 = 37
Award B1ft for correct Check in main answer
[M1] [A1] [SC1] [M1] [M2] [M1]

	Heat output (kW)	Dimensions of log burner (mm) Height	Width	Depth
Aston	3.75	510	389	291
Brunel	4	524	385	290
Chester	6	592	553	397
Dover	11	678	759	403

†3 (a) How many **centimetres** wider is the Dover than the Aston?

[3 marks]

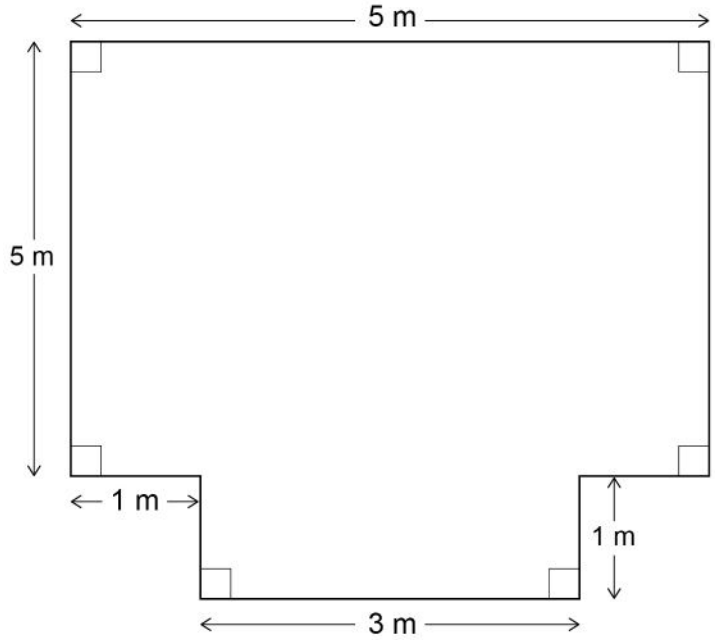
Check your answer.
Show how you have done your check.

[1 mark]

Do not write outside the box

3 (b) The height of Rohan's living room is 3 metres.

Here is a floor plan of the room.



Not drawn accurately

Rohan uses this formula to work out the heat output needed for the room.

$$K = \frac{A \times h}{14}$$

K is the heat output in kilowatts (kW)

A is the area of the floor in square metres (m^2)

h is the height of the room in metres (m)

What is the smallest log burner with enough heat output for this room?
You **must** show your working.

[5 marks]

Do not write outside the box

3 (c) Dania has a log burner.

The log burner can burn smokeless fuel or wood.

The table shows information about the amount used per hour and the cost.

	Smokeless Fuel	**Wood**
Amount used	0.8 kg per hour	1.5 kg per hour
Cost	£4.40 for an 8 kg bag	£5.40 for a 10 kg bag

Dania uses her log burner for

4 hours per day

5 days per week.

She says,

"Using smokeless fuel is cheaper than using wood by **more than** £7 per week."

Is she correct?

You **must** show your working.

[7 marks]

16

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4 **Apricot jam**

Here is a recipe for apricot jam.

Ingredients for one batch of jam	
Apricots	1.8 kg
Sugar	700 g
Water	125 ml

Method

Put the ingredients into a pan

Heat the mixture to 105 °C

Cook for 40 minutes

Q4(a) Worked Solution:
Alternative method 1
7.2 ÷ 1.8 or 4
their 4 × 700 or 700 ÷ 1000 or 2800 0.7
their 2800 ÷ 1000 or 2.8
their 0.7 × their 4
[M1] [M1] [M1]

4 (a) How many batches of jam can be made using 2 litres of water?

Circle your answer.

[1 mark]

0.8 1.6 8 16

4 (b) Peter is going to use this recipe to make apricot jam.

He has

7.2 kg of apricots

2.5 kg of sugar.

He says,

"I have enough sugar to make the jam using **all** 7.2 kg of apricots."

Is he correct?

You **must** show your working.

[5 marks]

Do not write outside the box

4 (c) Peter buys more apricots and sugar.

Altogether, he makes $17\frac{3}{4}$ pounds of jam.

1 pound = 16 ounces

He puts the jam into jars.

Each jar holds 11 ounces of jam.

How many jars can Peter fill with jam?

[3 marks]

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4 (d) Peter's costs for each jar of jam are

Apricots	£2.20
Sugar	11p
Jar	70p
Other costs	3p

Peter says,

"**More than** 70% of the cost is for apricots."

Is he correct?
You **must** show your working.

[4 marks]

13

END OF QUESTIONS

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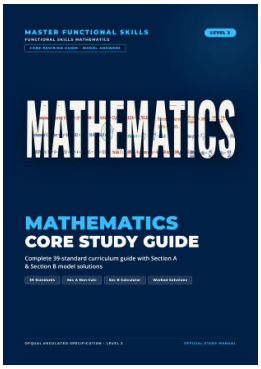
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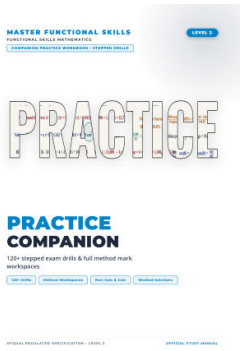


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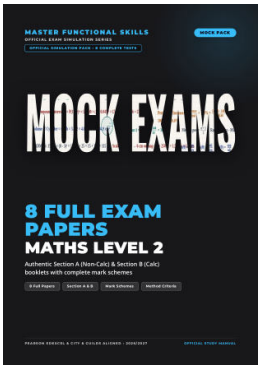


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